Lopcombe Corner — Chance of Reaching the 97 m Flood Threshold

Probability that the borehole level is at or above 97 mAOD (the Pitton groundwater-flood threshold) at some time later in the winter, given the level at the end of each month from October to February · EA E14150, 63 winters 1963/64–2025/26 · levels from the full EA record (manual dips and daily logger) · rainfall HadUK-Grid 1 km · 20 Oct 2026

In short. 7 of 63 winters reached 97 m (about 1 in 9): 1994/95, 2000/01, 2002/03, 2013/14, 2019/20, 2023/24, 2025/26. The October level tells you little — the well starts nearly every winter at about 78–79 m — so at the end of October the chance is roughly the long-run 1 in 10. From the end of November the level becomes a strong guide: a 50:50 chance corresponds to about **88–89 m at the end of November**, **92–93 m at the end of December**, **93–95.5 m at the end of January** and **about 96 m at the end of February**. Every flood winter was at least 89.6 m by the end of January, and no winter has first reached 97 m after the end of February. On 2026-10-20 the level was 78.25 m, an ordinary autumn level.

1. Probability table

Chance (%) that the level is at or above 97 m at some time after that month end (up to 31 May), by the level at the month end. Each cell gives the rainfall-scenario estimate, then the history estimate in brackets. A dash means that level has never been seen at that time of year, so no estimate is given.

Level at month end	End Oct	End Nov	End Dec	End Jan	End Feb
79 m	9 (14)	4 (7)	2 (5)	0 (0)	0 (0)
80 m	18 (15)	5 (9)	2 (6)	0 (0)	0 (0)
82 m	54 (19)	9 (16)	4 (9)	0 (0)	0 (0)
84 m	—	16 (25)	8 (14)	1 (1)	0 (0)
86 m	—	27 (37)	13 (21)	2 (3)	0 (0)
88 m	—	42 (51)	21 (30)	5 (7)	0 (1)
90 m	—	59 (65)	31 (41)	11 (17)	1 (2)
92 m	—	75 (77)	44 (52)	22 (36)	4 (7)
93 m	—	81 (82)	51 (58)	29 (48)	9 (12)
94 m	—	87 (86)	58 (64)	38 (60)	17 (21)
95 m	—	91 (89)	65 (69)	47 (71)	29 (33)
96 m	—	—	72 (74)	57 (80)	44 (48)

	End Oct	End Nov	End Dec	End Jan	End Feb
50:50 level (scenario)	82.0 m	89.0 m	93.0 m	95.5 m	96.5 m
50:50 level (history)	—	88.0 m	91.8 m	93.2 m	96.2 m
Highest level, no flood	82.8 m	87.6 m	93.2 m	93.7 m	94.2 m
Lowest level, then flooded	78.2 m	78.5 m	79.6 m	89.6 m	93.7 m

“Lowest level, then flooded”: the lowest month-end level from which the level later reached 97 m. 2013/14 was only 80.3 m at the end of December and peaked at 102.0 m in February; 2025/26 was 89.6 m at the end of January and reached 99.0 m; 2023/24 had fallen back to 93.7 m by the end of February and rose again to 99.1 m in April.

2. How to read it

- **End of October:** the level barely varies between winters (77.6–82.9 m), so it says little. Both methods put the chance near the long-run 1 in 10 at ordinary levels. Above about 80 m the two methods disagree sharply (see §4), so treat October estimates above 80 m as very uncertain.
- **End of November and December:** the most useful early warning. A rise to the high 80s by late November (as in 2000/01, 2002/03 and 2023/24) roughly halves the odds against a flood; but a low December level does not rule one out if January and February are very wet (2013/14).
- **End of January and February:** the curves steepen — the outcome depends on whether the level is already within a few metres of 97 m. Only three winters first crossed 97 m after the end of January (1994/95, 2019/20, 2025/26), none after the end of February. Late-season exceedances came from winters already near or above 97 m, such as 2023/24's return to 99.1 m in April 2024.
- Probabilities above the 97 m line are 100% by definition: the threshold has already been reached.

3. Methods

- **Data.** All manual dips (1963 onwards) and the daily 09:00 logger reading (1999 onwards) from the EA Hydrology API, downloaded 20 Oct 2026 (scripts/export_levels.py). One spurious dip (105.08 m, 15 May 2025) was removed. The month-end level is interpolated between the readings either side, which must be within 21 days. The outcome for each month is whether any reading after that month end, up to 31 May, is at or above 97 m.
- **History method.** For each month, a logistic regression of “reached 97 m later” on the month-end level, using Firth's penalised estimator because there are only seven flood winters. It answers: among past winters at this level on this date, how often did a flood follow?
- **Rainfall-scenario method.** For each month, a regression of the highest level after the month end on the month-end level and the HadUK-Grid rainfall still to come (from the next month to April). For a given level, the peak is then predicted under each of the 62 historical rainfall sequences, with the model's residual scatter, and the chances averaged. It answers: starting from this level, what fraction of plausible rest-of-winter weather leads to 97 m?
- Curves are drawn only across the range of levels seen at that time of year (plus a small margin).

Month end	Winters	Level coef (m per m)	Rain coef (m per 100 mm)	Fit R^2	Scatter (m)
Oct	58	2.75 ± 0.37	4.0 ± 0.3	0.80	2.8
Nov	60	0.96 ± 0.12	3.6 ± 0.4	0.77	3.0
Dec	59	0.75 ± 0.10	4.0 ± 0.6	0.70	3.3
Jan	60	0.79 ± 0.07	2.8 ± 0.7	0.74	3.0
Feb	62	0.90 ± 0.05	1.8 ± 0.5	0.85	2.2

Scenario-model regression for each month end. The scenario fit excludes 2025/26, because HadUK-Grid rainfall currently ends in December 2025.

4. Checks and caveats

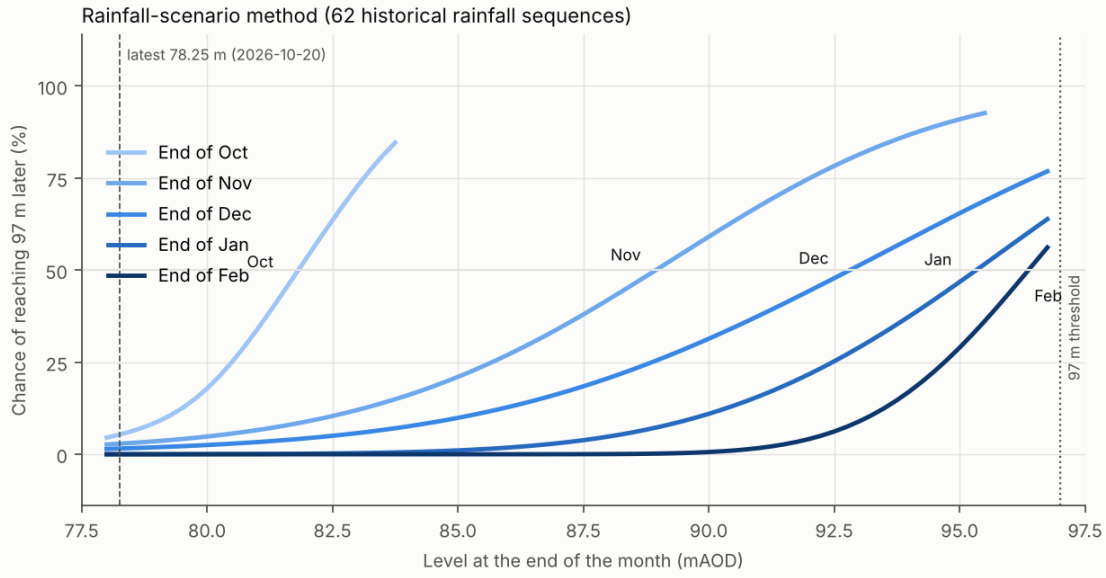
- **Calibration.** Added over all winters, the history method expects 7.3 flood winters against 7 observed (end-December case). The rainfall-scenario method expects 4.9 against 6 in the winters it covers, so it runs about 20% low — read it as the lower and the history method as the upper estimate.
- **The two methods agree well from November to February** (50:50 levels within about 2 m). In October they diverge above 80 m: the scenario model extrapolates a steep level effect, while history has only three winters above 81 m at the end of October (1966/67, 1993/94, 2024/25) — all peaked at 94–95 m but none reached 97 m.
- **Only seven flood winters.** Individual percentages carry wide uncertainty — think of them as ±10–20 points.
- **Rainfall scenarios** treat the rest of the winter's rain as unrelated to what has fallen so far, and give every historical winter equal weight; any shift toward wetter winters (the Oct–Mar trend is +1.4 mm/yr, not significant) is not included.
- **Before 1999** the record is manual dips (roughly weekly), so early peaks and month-end levels can be slightly low or smoothed.

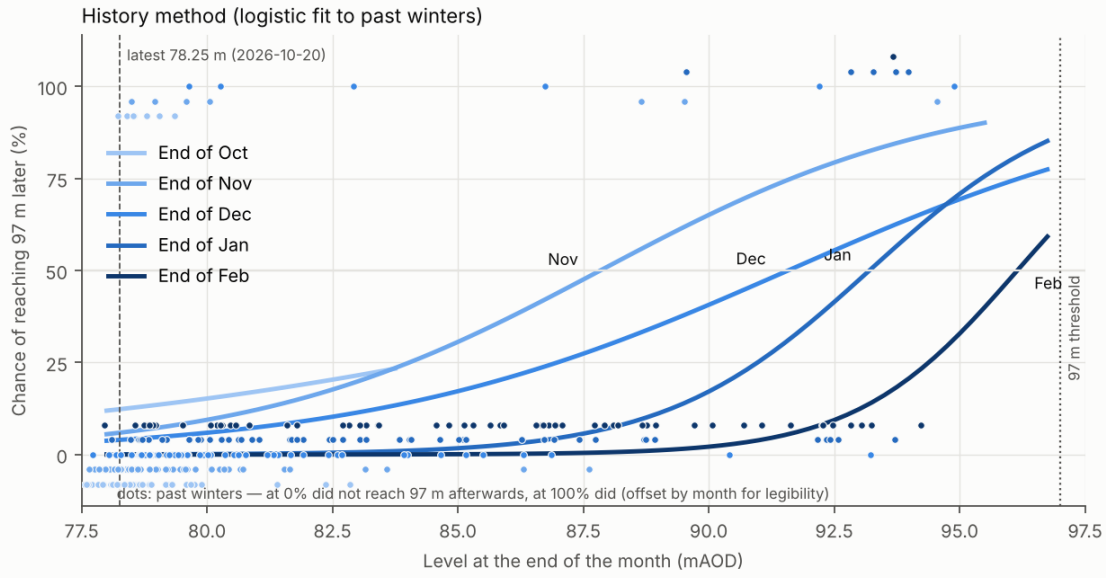
5. Data note — earlier winter peaks

The full daily logger record shows that several winter peaks used in the earlier HadUK-Grid, start-of-winter and Oct–Mar reports were too low, because the earlier extraction still missed later peaks. Notable differences: 2023/24 99.11 m (12 Apr 2024, not 97.54 m); 2024/25 95.04 m (not 90.31 m); 2014/15 87.78 m (not 83.13 m); 2021/22 82.68 m (not 79.61 m); 2012/13 95.73 m (not 94.42 m). Winter 2025/26 (99.02 m, 27 Feb 2026) is new. This report uses the full record, and the HadUK-Grid, start-of-winter, annual-maximum and Oct–Mar reports have since been re-run with it.

Reproduce: `python scripts/export_levels.py` then `python scripts/flood_probability.py` (outputs in output/flood/).

Lopcombe Corner — chance the level is at or above 97 m later in the winter, given the level at each month end (63 winters)





Appendix — month-end levels and winter peaks (mAOD)

Winter	End Oct	End Nov	End Dec	End Jan	End Feb	Peak	Peak date
1963/64	78.60	80.08	84.65	81.91	80.46	84.76	1964-01-01
1964/65	78.08	77.94	78.18	78.69	79.52	80.18	1965-04-30
1965/66	78.24	78.47	80.36	83.04	91.06	93.73	1966-03-05
1966/67	82.85	87.61	86.31	87.74	90.08	94.44	1967-03-26
1967/68	—	—	—	—	86.96	87.78	1968-02-08
1968/69	79.58	80.43	82.61	92.17	89.71	92.38	1969-02-05
1969/70	78.24	78.28	78.86	80.90	85.84	87.01	1970-04-02
1970/71	78.43	79.01	81.42	85.10	88.76	89.48	1971-02-19
1971/72	78.46	78.39	78.63	79.19	82.83	92.30	1972-03-26
1972/73	78.21	78.29	80.53	81.65	81.58	81.97	1973-02-12
1973/74	—	—	78.91	80.65	90.70	90.85	1974-03-05
1974/75	79.90	83.57	86.87	—	92.26	95.12	1975-02-12
1975/76	78.13	78.00	78.19	78.10	77.96	78.26	1975-10-01
1976/77	77.65	78.41	79.51	81.69	86.67	89.05	1977-03-12
1977/78	78.69	79.45	81.33	84.03	88.10	89.37	1978-03-19
1978/79	78.24	78.06	78.85	80.31	85.30	90.22	1979-04-23
1979/80	78.54	78.47	79.62	84.07	87.92	89.69	1980-04-16
1980/81	79.22	80.25	80.63	80.63	80.30	87.71	1981-04-23
1981/82	79.74	80.65	81.91	88.70	88.67	92.84	1982-03-31
1982/83	78.47	79.60	83.99	88.92	87.07	89.07	1983-01-29
1983/84	78.39	78.32	78.49	79.66	83.40	83.85	1984-03-09
1984/85	77.93	78.39	79.60	82.44	86.80	86.84	1985-02-26
1985/86	78.47	78.29	78.99	84.63	86.71	87.11	1986-02-19
1986/87	78.51	79.31	83.92	86.71	83.42	89.41	1987-04-27
1987/88	78.40	79.20	79.88	80.68	87.72	87.92	1988-03-04
1988/89	78.69	79.17	79.03	78.66	78.79	82.78	1989-04-30
1989/90	77.67	77.75	78.28	79.87	93.25	93.30	1990-02-28
1990/91	77.70	77.65	77.73	78.47	78.97	80.06	1991-04-30
1991/92	78.24	78.50	78.66	78.64	78.56	79.24	1992-05-22
1992/93	78.13	78.44	—	86.27	85.64	87.02	1993-02-12
1993/94	81.39	81.53	85.50	93.71	92.47	94.20	1994-01-28
1994/95	**78.54**	**80.06**	**82.93**	**93.98**	**98.19**	**98.26**	**1995-02-23**
1995/96	—	78.28	79.36	81.79	83.18	84.75	1996-03-19
1996/97	77.94	78.06	78.67	78.68	78.74	79.84	1997-04-02
1997/98	77.56	77.97	79.19	83.83	83.05	84.38	1998-02-05
1998/99	—	80.71	82.43	92.39	88.94	92.59	1999-02-03
1999/00	78.83	79.53	81.67	87.41	85.92	90.82	2000-05-13
2000/01	**79.35**	**89.52**	**98.03**	**93.28**	**97.55**	**101.39**	**2000-12-17**
2001/02	78.83	79.57	80.06	79.81	84.81	85.90	2002-04-03
2002/03	**78.80**	**94.55**	**94.89**	**97.53**	**93.05**	**100.75**	**2003-01-08**
2003/04	78.11	77.97	78.76	80.43	82.70	82.85	2004-03-05
2004/05	78.12	78.98	79.48	80.05	80.19	80.32	2005-02-13
2005/06	77.63	78.21	78.69	78.83	78.92	79.37	2006-05-01
2006/07	77.60	78.19	79.43	83.17	84.57	90.76	2007-03-19
2007/08	78.66	78.68	79.92	85.01	85.12	85.85	2008-02-09
2008/09	79.17	80.34	81.20	82.50	92.27	92.60	2009-02-23
2009/10	77.93	78.16	80.77	85.15	86.56	89.00	2010-04-20
2010/11	78.39	78.58	—	79.64	80.58	81.06	2011-03-25
2011/12	77.84	77.85	78.05	78.71	78.85	79.08	2012-05-31
2012/13	79.68	83.15	93.23	92.35	94.23	95.73	2013-02-11
2013/14	**78.40**	**78.96**	**80.26**	**97.81**	**100.95**	**102.01**	**2014-02-13**
2014/15	78.86	80.08	82.69	86.91	86.57	87.78	2015-02-13
2015/16	78.70	78.85	79.24	82.70	88.18	88.39	2016-03-04
2016/17	78.46	78.48	78.86	79.10	80.07	81.18	2017-04-09
2017/18	78.34	78.50	78.85	80.28	81.79	90.31	2018-04-25
2018/19	78.21	78.23	78.84	79.64	80.26	81.46	2019-04-24

Winter	End Oct	End Nov	End Dec	End Jan	End Feb	Peak	Peak date
2019/20	**78.53**	**79.59**	**86.75**	**92.83**	**97.01**	**97.25**	**2020-02-25**
2020/21	79.30	81.65	85.17	88.74	91.64	92.33	2021-02-20
2021/22	78.89	79.77	79.85	81.04	80.85	82.68	2022-04-07
2022/23	77.91	78.62	80.01	92.58	86.93	93.09	2023-01-26
2023/24	**79.05**	**88.65**	**92.19**	**93.72**	**93.67**	**99.11**	**2024-04-12**
2024/25	82.36	86.31	90.41	92.42	92.83	95.04	2025-02-11
2025/26	**78.23**	**78.49**	**79.64**	**89.56**	**98.87**	**99.02**	**2026-02-27**

Bold: reached 97 m. “—”: no readings within 21 days of that month end.